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1

If $A = \begin{bmatrix} 1 & -3 \\ 6 & 9 \end{bmatrix}$, then $(3A)^{-1} = \underline{\hspace{2cm}}$

A) $\frac{1}{81} \begin{bmatrix} 9 & 3 \\ -6 & 1 \end{bmatrix}$

B) $\frac{1}{249} \begin{bmatrix} 27 & -9 \\ 18 & 3 \end{bmatrix}$

C) $\begin{bmatrix} \frac{1}{9} & \frac{1}{27} \\ \frac{2}{27} & \frac{1}{81} \end{bmatrix}$

D) None

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2

If $\begin{bmatrix} 1 & -3 \\ 2 & 4 \end{bmatrix} X = \begin{bmatrix} 4 & -12 \\ 8 & 16 \end{bmatrix}$, then $X =$ _____

A) $\frac{1}{4}$ B) 4 C) $4I$ D) $\frac{I}{4}$

Jagrut Awaaz

3

If Linear Equation $2x + 3y = 2xy$, $8x + 3y = 5xy$, then $\begin{bmatrix} y \\ x \end{bmatrix} = \underline{\hspace{2cm}}$

- A) $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ B) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ C) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ D) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

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If $A = \begin{bmatrix} \tan \theta & \sec \theta \\ \sec \theta & \tan \theta \end{bmatrix}$, then $(A)^{-1} = \underline{\hspace{2cm}}$

A) $-\sec \theta \begin{bmatrix} \sin \theta & -1 \\ -1 & \sin \theta \end{bmatrix}$

B) $\begin{bmatrix} \tan \theta & -\sec \theta \\ -\sec \theta & \tan \theta \end{bmatrix}$

C) $\begin{bmatrix} \tan \theta & \sec \theta \\ \sec \theta & \tan \theta \end{bmatrix}$

D) None

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5

Period of $4 \sin\left(\frac{x}{2}\right) + 7 \cot\left(\frac{x}{10}\right) = \underline{\hspace{2cm}}$

- A) 14π B) 20π C) 40π D) 86π

Jagrut Awaaz

6

$$\frac{\sin(\theta - \pi)}{\cos(\theta - \frac{\pi}{2})} \times \frac{\tan(\theta - \frac{3\pi}{2})}{\cot(\theta - 2\pi)} = \underline{\hspace{2cm}}$$

- A) 1 B) 0 C) $\sin^2 \theta$ D) -1

Jagrut Awaaz

. If $\bar{a} + \bar{b} + \bar{c} = \bar{0}$, then $\bar{b} \times \bar{c} = \underline{\hspace{2cm}}$

- A) $\bar{c} \times \bar{b}$ B) $-(\bar{c} \times \bar{a})$ C) $\bar{b} \times \bar{a}$ D) $\bar{a} \times \bar{b}$

Jagrut Awaaz

8

$$i \wedge j = \underline{\hspace{2cm}}$$

A) $\frac{\pi}{3}$ B) $\frac{\pi}{2}$ C) $\frac{\pi}{6}$ D) 0

Jagrut Awaaz

Equation of line passing through points $\left(\frac{b}{a}, \frac{a}{b}\right)$ and slope $\frac{-a}{b}$ is = ____

A) $ax + by - a + b = 0$

B) $ax + by + a - b = 0$

C) $ax + by - a - b = 0$

D) $ax + by + a + b = 0$

Jagrut Awaaz

The equation of circle whose centre is $(3, 2)$ and touching x - axis is _____

A) $x^2 + y^2 - 6x - 4y + 9 = 0$ B) $x^2 + y^2 - 6x - 4y + 4 = 0$

C) $x^2 + y^2 - 6x - 4y - 9 = 0$ D) $x^2 + y^2 - 6x - 4y - 4 = 0$

Jagrut Awaaz

$$\lim_{x \rightarrow 0} \frac{x^2 \cdot \cot 6x}{6^{3x+1} - 6} = \underline{\hspace{2cm}}$$

A) $\frac{1}{36 \log_e 6}$

B) $\frac{\log_6 e}{108}$

C) $\frac{1}{3 \log_e 6}$

D) Does not exist



$$\lim_{n \rightarrow \infty} \frac{n \cdot \sum n^2}{\sum n^3} = \underline{\hspace{2cm}}$$

A) $\frac{4}{3}$

B) $\frac{1}{2}$

C) $\frac{3}{4}$

D) 1

Jagrut Awaaz

. If $f(x) = \frac{x}{1+x}$, then $f(y) + f\left(\frac{1}{y}\right) = \underline{\hspace{2cm}}$

A) $2f(y)$

B) 0

C) 2

D) 1

Jagrut Awaaz

. If $x = \frac{1}{1+t^2}$, $y = \frac{2t}{1+t^2}$, then $\frac{dy}{dx} =$ _____

A) $\frac{t^2}{1+t^2}$ B) $\frac{1-t^2}{t}$ C) $\frac{t^2-1}{t}$ D) $\frac{t}{1+t}$



If $y = 2e^{3x} + 3e^{-2x}$, then $\frac{d^2y}{dx^2} - \frac{dy}{dx} = \underline{\hspace{2cm}}$

A) $4y$ B) $2y$ C) $-6y$ D) $6y$

Jagrut Awaaz



If $y = 7x^9 + 3x^2 + 4e^{2x}$, then $\frac{d^{20}y}{dx^{20}} = \underline{\hspace{2cm}}$

A) $4e^{2x} \cdot 2^{20}$ B) $e^{2x} \cdot 2^{40}$ C) e^{2x} D) 0

Jagrut Awaaz

$$\frac{d^2}{dx^2} [3^x(\tan^{-1} x + \cot^{-1} x)] = \underline{\hspace{2cm}}$$

A) 3^x B) 0 C) $3^x \cdot (\log 3)^2 \frac{\pi}{2}$ D) $\frac{\pi}{2}$

Jagrut Awaaz

$$\cdot \int \frac{1}{e^x + e^{-x}} dx = \underline{\hspace{2cm}} + c$$

A) $e^x - e^{-x}$

B) $\tan^{-1} e^x$

C) $e^x + e^{-x}$

D) Not Possible

Jagrut Awaaz

$$\int \frac{e^{\tan^{-1} x}}{1+x^2} dx = \text{_____} + c$$

A) $\tan^{-1} x$ B) $e^{\tan^{-1} x}$ C) e D) Not Possible

Jagrut Awaaz



$$\int_{-\pi}^{\pi} x^{121} dx = \underline{\hspace{2cm}}$$

A) $\frac{\pi^{122}}{122}$

B) $\frac{\pi^{122}}{61}$

C) 1

D) 0

Jagrut Awaaz

$$\int_0^{\frac{\pi}{2}} \frac{2(\sqrt[3]{\sec x})}{\sqrt[3]{\sec x} + \sqrt[3]{\operatorname{cosec} x}} dx = \underline{\hspace{2cm}}$$

A) $\frac{\pi}{4}$ B) 0 C) 2 D) $\frac{\pi}{2}$

Jagrut Awaaz

, $3^{\log_x 4} = 2$, then $x =$ _____

- A) 9 B) 3 C) $\sqrt{3}$ D) 81

Jagrut Awaaz

$$\log_y x \times \log_z y \times \log_w z \times \log_w p = \underline{\hspace{2cm}}$$

A) 0 B) $\log \frac{xp}{w^2}$ C) $\log_w x \times \log_w p$ D) $\log_e x \times \log_{w^2} p$

Jagrut Awaaz

Mean of first six two digit prime number is = _____

A) $\frac{56}{3}$

B) $\frac{29}{6}$

C) $\frac{41}{6}$

D) $\frac{52}{3}$

Jagrut Awaaz



$$\frac{\sin 79^\circ + \sin 19^\circ}{\cos 19^\circ - \cos 79^\circ} = \underline{\hspace{2cm}}$$

A) $\tan \frac{\pi}{4}$

B) $\tan \frac{\pi}{2}$

C) $\tan \frac{\pi}{6}$

D) $\tan \frac{\pi}{3}$

Jagrut Awaaz

If $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -6 \\ 7 & -9 \end{bmatrix}$ then $(A - 2B)^{-1} = \underline{\hspace{2cm}}$

- A) $\frac{1}{56} \begin{bmatrix} 14 & 14 \\ -11 & 7 \end{bmatrix}$ B) $\frac{1}{56} \begin{bmatrix} 14 & -14 \\ 11 & -7 \end{bmatrix}$ C) $\frac{1}{56} \begin{bmatrix} 14 & -14 \\ -11 & -7 \end{bmatrix}$ D) $\frac{1}{56} \begin{bmatrix} 14 & 14 \\ 11 & 7 \end{bmatrix}$

Jagrut Awaaz

. $\begin{bmatrix} 0 & 10 & 20 \\ -10 & 0 & -3 \\ -20 & 3 & 0 \end{bmatrix}$ is _____ *matrix* (શ્રેણિક)

A) Skew-symmetric (વિસંમિત)

B) Symmetric Matrix (સંમિત)

C) Diagonal (વિકર્ણ)

D) Zero (શૂન્ય)

$$\begin{bmatrix} 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 6 & -2 \\ 1 & 3 \\ 4 & -6 \end{bmatrix} \begin{bmatrix} 9 \\ -8 \end{bmatrix} = \underline{\hspace{2cm}}$$

- A) $[3]$ B) $[361]$ C) $\begin{bmatrix} 3 \\ 361 \end{bmatrix}$ D) Not possible (શક્ય નથી)

If $\begin{bmatrix} x+y & 3 \\ x-y & 4 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, then $\frac{x}{y} = \underline{\hspace{2cm}}$

A) $-\frac{1}{3}$ B) $-\frac{1}{2}$ C) $\frac{3}{2}$ D) -3

Jagrut Awaaz

The principle Period (અવધાન) of $3 \sin\left(4 + \frac{\pi\theta}{2}\right) + 6$ is _____

- A) $\sin^2\left(\frac{\pi}{6}\right)$ B) $\tan^2\left(\frac{\pi}{4}\right)$ C) $\operatorname{cosec}^2\left(\frac{\pi}{6}\right)$ D) $\cot^2\left(\frac{\pi}{6}\right)$

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$$\cos 15^\circ + \sin 15^\circ = \underline{\hspace{2cm}}$$

A) $\frac{\sqrt{6} + \sqrt{2}}{4}$

B) $\frac{\sqrt{6} - \sqrt{2}}{4}$

C) $\frac{\sqrt{6}}{5}$

D) $\sqrt{\frac{3}{2}}$

Jagrut Awaaz

$$\frac{\cos 3A + 2\cos 5A + \cos 7A}{\sin 3A + 2\sin 5A + \sin 7A} = \underline{\hspace{2cm}}$$

- A) $\cot 7A$ B) $\cot 5A$ C) $\tan 4A$ D) $\tan 5A$

Jagrut Awaaz

If vectors $\vec{a} = (6, 0, -7)$, $\vec{b} = (1, -1, 3)$ and $\vec{c} = (2, -7, -9)$, then $\vec{a} \cdot (\vec{b} \times \vec{c}) = \underline{\hspace{2cm}}$
A) 45 B) 215 C) 315 D) 95

Jagrut Awaaz

Angle between line $x - \sqrt{3}y + 2 = 0$ and $\sqrt{3}x - y + 1 = 0$ is = ____

- A) $\frac{\pi}{3}$ B) $\frac{\pi}{6}$ C) $\frac{\pi}{2}$ D) $\frac{\pi}{4}$

Jagrut Awaaz

Angle between two vectors $3j - 2i + k$ and $i + 2j + k$ is = _____

A) $\cos^{-1} \sqrt{\frac{59}{84}}$

B) $\sin \sqrt{\frac{5}{2\sqrt{21}}}$

C) $\cos^{-1} \sqrt{\frac{5}{4\sqrt{21}}}$

D) $\sin^{-1} \sqrt{\frac{59}{84}}$

Jagrut Awaaz

.. The equation of a circle passing through points $(4, 0)$, $(0, 4)$, $(0, 0)$ is _____

$\{(4, 0), (0, 4), (0, 0)$ બિંદુઓમાંથી પસાર થતા વર્તુળનું સમીકરણ _____}

A) $x^2 + y^2 - 4x + y = 0$

B) $2x^2 + 2y^2 - 4x - 3y = 0$

C) $x^2 + y^2 - 4x - 4y = 0$

D) $x^2 + y^2 - 4x$



$$\lim_{x \rightarrow \infty} \left[\left(\frac{4+x}{3+x} \right)^x + \frac{3x^2+4}{3-x^2} \right] = \underline{\hspace{2cm}}$$

A) $e - 3$ B) $e^4 - e^3$

C) 1

D) 0

Jagrut Awaaz

$$\cdot \lim_{\theta \rightarrow 0} \frac{\sin^2 \pi \theta}{2\theta^2} = \underline{\hspace{2cm}}$$

A) $\frac{\pi}{2}$

B) $\frac{\pi^2}{2}$

C) $\frac{\pi}{4}$

D) $\frac{\pi^2}{4}$

Jagrut Awaaz

If $f(x) = \sin \frac{x}{2}$, then $f(\pi) - f\left(\frac{\pi}{2}\right) =$ _____

A) $\frac{\sqrt{2}-1}{\sqrt{2}}$

B) $\frac{-1}{\sqrt{2}}$

C) $\frac{1}{\sqrt{2}}$

D) $\frac{\sqrt{2}+1}{\sqrt{2}}$

Jagrut Awaaz



$$\frac{d}{dt} \left(\frac{1-t^2}{1+t^2} \right) = \underline{\hspace{2cm}}$$

A) $\frac{1+t^2}{1-t^2}$

B) $\frac{1-t^2}{(1+t^2)^2}$

C) $\frac{-4t}{(1+t^2)^2}$

D) 0

Jagrut Awaaz

If $x^3 + y^3 = x^3 y^3$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$

A) $\frac{3x^2 - y^3}{y^2 - 3x^2}$

B) $\frac{x^3 + y^3}{x^2 - y^2}$

C) $\frac{(x^2 y^3 - x^2)}{(y^2 - x^3 y^2)}$

D) $\frac{x^4}{y^4}$

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. The equation of a motion of a particle (કણની ગતિનું સમીકરણ) is $S = t^3 - 6t^2 + 9t + 4$, when $a = 0$ then $v = \text{---} \text{m/s}$
A) -2 B) 6 C) 4 D) -3

Jagrut Awaaz



If $y = 4e^{3x} + 2^x$ then $\frac{d^3y}{dx^3} = \underline{\hspace{2cm}}$

- A) $108 e^{3x} + 2^x(\log 2)^3$ B) $4^3 \cdot e^{3x} + 2^x$ C) $4^3 \cdot e^{3x} + 2^x(\log 2)^3$ D) Not possible

Jagrut Awaaz

$$\int_0^{\frac{\pi}{2}} \sin^4 x \cdot \cos x \, dx = \underline{\hspace{2cm}}$$

A) $-\frac{1}{5}$ B) $\frac{1}{5}$ C) 1 D) 0

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$$\int \frac{4 + 3\cos x}{\sin^2 x} dx = \text{_____} + C$$

- A) $4 \cot x + 3 \operatorname{cosec} x$ B) $-4 \cot x + 3 \operatorname{cosec} x$ C) $4 \cot x - 3 \operatorname{cosec} x$ D) *None of the above*

Jagrut Awaaz



$$\int x \, dy = \underline{\hspace{2cm}} + C$$

A) $\frac{x^2}{2}$

B) 0

c) $x \cdot y$

D) 1

Jagrut Awaaz



$$\int_{-\pi}^{\pi} x^2 \cdot \sin x \, dx = \underline{\hspace{2cm}}$$

- A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) $\frac{\pi}{2}$ D) 0

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$$\log_{25} 100 - \log_2 16 = \underline{\hspace{2cm}}$$

- A) $\log_5 10 - 4$ B) $\log_{10} 5 - 4$ C) $\log_5 \frac{125}{2}$ D) $\log_5 10 - 2$

Jagrut Awaaz

If $\log_2(x + 3) + \log_2(x - 3) = 1$, then $x =$ ____

- A) -7 B) $\sqrt{11}$ C) 1 D) 0

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The Mean (अवस्था) of 2, 3, 7, 8, 9, 11, 14, 10 is _____

- A) 5 B) -2 C) 8 D) -8

Jagrut Awaaz

If $[0 \ x \ -2] \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = [4]$, then $x =$ _____ .

A. 2

B. 4

C. 5

D. 6

If A is a square matrix then, $A - A^T$ is _____ matrix.

- A. Diagonal matrix B. Symmetric C. Skew-symmetric D. Row

Jagrut Awaaz

If $\begin{bmatrix} x & 3 \\ y & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \end{bmatrix}$, then $y =$ _____ .

A. 4

B. 9

C. 3

D. 2

Jagrut Awaaz

If $A = \begin{bmatrix} 1 & 4 \\ 3 & -2 \end{bmatrix}$, then $2A - 3I =$ _____.

A. $\begin{bmatrix} 1 & 8 \\ 6 & -4 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 8 \\ 6 & 7 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 8 \\ 6 & -7 \end{bmatrix}$

D. $\begin{bmatrix} 1 & -8 \\ -6 & 7 \end{bmatrix}$

If $\tan \alpha = \frac{1}{2}$ and $\tan \beta = \frac{1}{3}$ then, $\tan (\alpha + \beta) = \underline{\hspace{2cm}}$.

- A. 1 B. $\frac{1}{\sqrt{2}}$ C. $\frac{\sqrt{3}}{2}$ D. 0

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$$\sin^2 40^\circ + \sin^2 50^\circ = \underline{\hspace{2cm}}$$

A. 1

B. -1

C. 0

D. None of these

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If $\sin\theta = \frac{3}{5}$, then $\cos 2\theta =$ _____ .

A. $\frac{16}{25}$

B. $\frac{7}{25}$

C. $\frac{25}{7}$

D. $\frac{4}{5}$

Jagrut Awaaz

Force acting on the particle F and displacement of the particle is d , then work done $W =$ _____ .

A. $F \times d$

B. $d \times F$

C. $F \cdot d$

D. None of above

If $a = i - j + k$, $b = 2i - j + k$ and $c = i + j - 2k$ then find $a \cdot (b + c)$.

A. 2

B. 4

C. 3

D. 1

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Equation of line passing through points (5, 8) and (4, 3) is

- A. $5x + y + 17 = 0$ B. $5x - y - 17 = 0$ C. $-5x + y + 17 = 0$ D. $5x - y + 17 = 0$

Jagrut Awaaz

Centre of circle $x^2 + y^2 + 3x - 4y - 4 = 0$ is

- A. $\left(\frac{-3}{2}, 2\right)$ B. $\left(\frac{3}{2}, -2\right)$ C. $\left(\frac{-3}{2}, -2\right)$ D. $\left(\frac{3}{2}, 2\right)$

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If $f(x) = \log \left(\frac{x-1}{x} \right)$, then $f(-x) = \underline{\hspace{2cm}}$.

A. $\log \left(\frac{x-1}{x} \right)$

B. $\log \left(\frac{x+1}{x} \right)$

C. $\log \left(\frac{x}{x+1} \right)$

D. $\log \left(\frac{x}{x-1} \right)$

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 7}{x^2 + 3x + 1} = \underline{\hspace{2cm}}.$$

A. 1

B. -3

C. 3

D. -1

A solid yellow square is located in the top-left corner of the page.
$$\lim_{x \rightarrow 0} (\sec^2 x - \tan^2 x) = \underline{\hspace{2cm}} .$$

A. -1

B. 0

C. 1

D. 5

$\frac{d}{dx} (\tan^{-1}x + \cot^{-1}x) = \underline{\hspace{2cm}} .$

A. 1

B. -1

C. 0

D. $\frac{\pi}{2}$

If the equation of motion of a particle is $S = 3t^2 - 1$, then velocity at $t = 2$ seconds is

A. 12 m/s B. 18 m/s C. 6 m/s D. None of these

Jagrut Awaaz

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \underline{\hspace{2cm}} .$$

A. $\frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2}$

B. $\frac{v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx}}{v^2}$

C. $\frac{u \cdot \frac{dv}{dx} - v \cdot \frac{du}{dx}}{v^2}$

D. $\frac{u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx}}{v^2}$

$$\frac{d}{dx} \tan x = \underline{\hspace{2cm}} .$$

A. $\sec^2 x$

B. $-\sec^2 x$

C. $\sec x \cdot \tan x$

D. $-\sec x \cdot \tan x$

$$\int e^{2x} dx = \underline{\hspace{2cm}} .$$

A. $e^{2x} + c$

B. $2x + c$

C. $\frac{e^{2x}}{2} + c$

D. e^{2x}

$$\int x \cdot e^x dx = \underline{\hspace{2cm}} .$$

A. e^x

B. $x \cdot e^x$

C. $(x - 1) e^x$

D. $(x + 1) e^x$

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$$\int_0^{2\pi} 1 \, dx = \underline{\hspace{2cm}} .$$

A. 1

B. 0

C. π D. 2π

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$$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx = \underline{\hspace{2cm}}.$$

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

$$\log_2 \left(\frac{1}{8} \right) = \underline{\hspace{2cm}}$$

A. 2

B. $\frac{-1}{2}$

C. -3

D. $\frac{1}{3}$

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$$(\log_b a) (\log_c b) (\log_a c) = \underline{\hspace{2cm}} .$$

A. 0

B. 1

C. abc

D. None of these

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The median of observations 37, 39, 47, 45, 41, 43 is _____

- A. 41 B. 42 C. 43 D. 45

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Most Imp MCQs

1

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then $A^2 + 2A = \dots\dots\dots$

જો $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ તો $A^2 + 2A = \dots\dots\dots$

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Most Imp MCQs

2

If A is square matrix then $A - A^T$ is :

જો A ચોરસ શ્રેણિક હોય તો $A - A^T$ એ.... શ્રેણિક છે.

Jagrut Awaaz



Most Imp MCQs

3

If $A^2 - 7A = I$ then $A^{-1} = \dots\dots\dots$

જો $A^2 - 7A = I$ તો $A^{-1} = \dots\dots\dots$

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4 $2 \sin^2\theta + 3\cos^2\theta = \dots\dots\dots$

Jagrut Awaaz



Most Imp MCQs

5

Peirod of $4 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ is :

$4 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ નું અવર્તમાન છે.

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6

$$\operatorname{cosec}(390^\circ) = \dots\dots\dots$$

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Most Imp MCQs

■ $\vec{a} = (2, -1, 1)$, $\vec{b} = (1, 1, -2)$,
 $\vec{c} = (1, 3, -1)$ If $\vec{a}$ is perpendicular to $(\lambda\vec{b} + \vec{c})$ then $\lambda =$
.....

જો $\vec{a} = (2, -1, 1)$, $\vec{b} = (1, 1, -2)$,
 $\vec{c} = (1, 3, -1)$ તથા $\vec{a}$ એ $(\lambda\vec{b} + \vec{c})$ ને લંબ હોય તો
 $\lambda =$

8

$$\mathbf{i} \cdot (\mathbf{j} \times \mathbf{k}) + \mathbf{j} \cdot (\mathbf{k} \times \mathbf{i}) + \mathbf{k} \cdot (\mathbf{i} \times \mathbf{j}) = \dots\dots\dots$$

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9

The slope of line joining the points $(1,3)$ and $(1,-5)$ is ...
બિંદુઓ $(1,3)$ અને $(1,-5)$ ને જોડતી રેખાનો ઢાળ

Jagrut Awaaz



Most Imp MCQs

10

The sum of x-intercept and y-intercept of line $4x + 3y = 12$ is

રેખા $4x + 3y = 12$ નાં x-અંતઃખંડ અને y-અંતઃખંડનો સરવાળો

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Most Imp MCQs




$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x} = \dots\dots\dots$$

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$$\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 3}{4x^2 + 4x + 7} = \dots\dots\dots$$



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13

$$\lim_{x \rightarrow 0} \frac{\tan x - 5x}{7x - \sin x} = \dots\dots\dots$$

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Most Imp MCQs

14

$$\frac{d}{dx} (e^{-\log_e x}) = \dots$$



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Most Imp MCQs

15

If $x = a \cos \theta$, $y = a \sin \theta$ then $\frac{d^2y}{dx^2} = \dots$

જો $x = a \cos \theta$, $y = a \sin \theta$ તો $\frac{d^2y}{dx^2} = \dots$

Jagrut Awaaz



Most Imp MCQs

■ If $\sqrt{x} + \sqrt{y} = \sqrt{a}$ then $\frac{dy}{dx} = \dots\dots\dots$

✂ If $\sqrt{x} + \sqrt{y} = \sqrt{a}$ then $\frac{dy}{dx} = \dots\dots\dots$

Jagrut Awaaaz

17

$$\int \frac{3x^2}{x^3 + 1} dx = \dots + C$$

Jagrut Awaaz

Most Imp MCQs

18

$$\int \frac{x^4 + 3x^2}{x^2 + 3} dx = \dots + C$$



Jagrut Awaaz

19

$$\int e^x(\tan x + \sec^2 x) dx = \dots + C$$

Jagrut Awaaz

20

$$\log_4 32 - \log_3 81 = \dots\dots\dots$$

Jagrut Awaaz



Most Imp MCQs

21

If $\log_2(x+3) + \log_2(x-3) = \log_2 7$ then $x = \dots$

જો $\log_2(x+3) + \log_2(x-3) = \log_2 7$ તો $x = \dots$

Jagrut Awaaz



Most Imp MCQs

22

If $\bar{x} - M = 4$ and $\bar{x} + M = 140$ then $M = \dots$

જો $\bar{x} - M = 4$ અને $\bar{x} + M = 140$ તો $M = \dots$

Jagrut Awaaz

23

$$\frac{d}{dx} \left(e^{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} \right) = \dots\dots\dots$$

Jagrut Awaaz

Most Imp MCQs

24

$$\lim_{x \rightarrow 0} \frac{e^{2x} - \sin 2x - 1}{2x} = \dots\dots\dots$$



Jagrut Awaaz

Most Imp MCQs

If $\begin{vmatrix} 1 & 2 & 3 \\ 2 & -x & -1 \\ 5 & 10 & 2 \end{vmatrix} = 0$ then $x = \dots\dots$

જલ $\begin{vmatrix} 1 & 2 & 3 \\ 2 & -x & -1 \\ 5 & 10 & 2 \end{vmatrix} = 0$ જલ $x = \dots\dots$

Jagrut Awaaz

$$\lim_{x \rightarrow 1} \frac{x^2 + 1}{x + 100} = \dots\dots$$



Jagrut Awaaz

Most Imp MCQs

27

$$\lim_{x \rightarrow 1} \frac{x^{15} - 1}{x - 1} = \dots\dots$$



Jagrut Awaaz

$$\lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 2x} = \dots\dots$$



Jagrut Awaaz

$$\lim_{x \rightarrow 0} \frac{1 - \cos p x}{1 - \cos q x} = \dots\dots$$



Jagrut Awaaz

Most Imp MCQs



$$\frac{d}{dx} \sqrt{\sin x} = \dots$$

Jagrut Awaaz



Most Imp MCQs

31

If $f(x) = x \sin x$ then $f' \left(\frac{\pi}{2} \right) = \dots\dots\dots$

જલ $f(x) = x \sin x$ ઢલ $f' \left(\frac{\pi}{2} \right) = \dots\dots\dots$

Jagrut Awaaz

32

$$\frac{d}{dx} (3 \sin 2 0^0 - 4 \sin^3 2 0^0) = \dots\dots\dots$$

Jagrut Awaaz

Most Imp MCQs

33

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1 + \cos 2x} dx = \dots\dots$$



Jagrut Awaaz

$$\int_0^{\frac{\pi}{2}} e^{\sin x} \cos x \, dx = \dots\dots$$



Jagrut Awaaz

$$\int \frac{1}{\sqrt{1-x}} dx = \dots + C$$



Jagrut Awaaz

$$\int \frac{\cos^4 x - \sin^4 x}{\cos^2 x - \sin^2 x} dx = \dots\dots + C$$



Jagrut Awaaz

Most Imp MCQs

37

If $\vec{a} = (l, m, n)$ and $\vec{b} = (m-n, n-l, l-m)$ then angle between $\vec{a}$ and $\vec{b}$ is :

જો $\vec{a} = (l, m, n)$ અને $\vec{b} = (m-n, n-l, l-m)$ તો $\vec{a}$ અને $\vec{b}$ વચ્ચેનો ખૂણો.....છે.

Jagrut Awaaz



Most Imp MCQs

38

Direction cosine of $\vec{a} = (6, -2, 3)$ is.....
 $\vec{a} = (6, -2, 3)$ ની દિક્કોસાઈન.....છે.

Jagrut Awaaz

39

$$2\log_6 2 + 3\log_6 3 + \log_6 12 = \dots\dots\dots$$

Jagrut Awaaz

40

$$6^{\log_6 5} + 3^{\log_9 16} = \dots\dots\dots$$

Jagrut Awaaz



Most Imp MCQs

41

If $\bar{x} = 25$ and $z = 20$ then $M = \dots\dots\dots$

જો $\bar{x} = 25$ અને $z = 20$ તો $M = \dots\dots\dots$

Jagrut Awaaz

Most Imp MCQs

42

If the centre of the circle $x^2 + y^2 + 5kx - 2y - 1 = 0$ is $(3,1)$ then $k = \dots$

જો વર્તુળ $x^2 + y^2 + 5kx - 2y - 1 = 0$ નું કેન્દ્ર છે $(3,1)$ હોય તો $k = \dots$

Jagrut Awaaz

43

The possible equation of the line perpendicular to the line $x + 2y + 2 = 0$ is

રેખા $x + 2y + 2 = 0$ ને લંબ રેખાનું શક્ય સમીકરણ છે.

Jagrut Awaaz

44

$$\cos^2(7.5^\circ) - \cos^2(37.5^\circ) = \dots$$

Jagrut Awaaz

45

$$\log (\cot 1^{\circ}) + \log (\cot 2^{\circ}) + \dots + \log (\cot 89^{\circ}) = \dots\dots\dots$$

Jagrut Awaaz

$$\cos(5A + 3B) + \cos(3A + B) = \dots$$



Jagrut Awaaz



Most Imp MCQs

47

$$\text{If } \begin{vmatrix} \sin x & \cos x \\ \cos x & \sin x \end{vmatrix} = 0 \text{ then } x = \dots$$
$$\text{or } \begin{vmatrix} \sin x & \cos x \\ \cos x & \sin x \end{vmatrix} = 0 \text{ then } x = \dots$$

Jagrut Awaaz



Most Imp MCQs

48

If $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ and $A^{-1} = \frac{1}{k} \text{adj}A$ then $k = \dots\dots$

જો $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ અને $A^{-1} = \frac{1}{k} \text{adj}A$ તો $k = \dots\dots$

Jagrut Awaaz

49

If $A = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix}$ then $\text{adj}A = \dots\dots$

જલ $A = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix}$ નલ $\text{adj}A = \dots\dots$

Jagrut Awaaz

Most Imp MCQs

Sum of minor and co-factor of 7 in $\begin{vmatrix} 2 & 3 & 5 \\ 1 & 0 & 7 \\ -1 & -2 & 4 \end{vmatrix}$ is

.....

$$\begin{vmatrix} 2 & 3 & 5 \\ 1 & 0 & 7 \\ -1 & -2 & 4 \end{vmatrix}$$

સરવાળો છે.



Most Imp MCQs

51

If $A = \begin{bmatrix} 5x & 10 \\ 8 & 7 \end{bmatrix}$ and $|A| = 25$ then $x = \dots$

જો $A = \begin{bmatrix} 5x & 10 \\ 8 & 7 \end{bmatrix}$ અને $|A| = 25$ તો $x = \dots$

Jagrut Awaaz

52

If $AB = X$ then
જલ $AB = X$ તલ

Jagrut Awaaz



Most Imp MCQs

53

If $A(x) = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$ then $A(\pi) = \dots\dots\dots$

જો $A(x) = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$ તો $A(\pi) = \dots\dots\dots$

Jagrut Awaaaz

Most Imp MCQs

54

If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ then $A (\text{adj}A) = \dots\dots$

જલ $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ તલ $A (\text{adj}A) = \dots\dots$

Jagrut Awaaz



Most Imp MCQs

55

If $f(x) = \tan^{-1}x$ then $f'(0) = \dots$

જો $f(x) = \tan^{-1}x$ તો $f'(0) = \dots$

Jagrut Awaaz

56

If $y = 100e^{2x} + 200e^{-2x}$ and $\frac{d^2y}{dx^2} = ky$ then $k = \dots\dots\dots$

જો $y = 100e^{2x} + 200e^{-2x}$ અને $\frac{d^2y}{dx^2} = ky$ તો $k = \dots\dots\dots$

Jagrut Awaaz

Most Imp MCQs

57

If $y = t^{10} + 1$ and $x = t^8 + 1$ then $\frac{dy}{dx} = \dots\dots\dots$

જો $y = t^{10} + 1$ અને $x = t^8 + 1$ તો $\frac{dy}{dx} = \dots\dots\dots$

Jagrut Awaaz

58

$$\frac{d}{dx} \log \left(\frac{e^x}{1 + e^x} \right) = \dots\dots\dots$$

Jagrut Awaaz

Most Imp MCQs

59

$$\lim_{n \rightarrow \infty} \frac{\sqrt{3n^2+5}}{5n+6} = \dots\dots\dots$$



Jagrut Awaaz

60

$$\int_0^{\pi} \cos^3 x \cdot \sin^4 x \, dx = \dots\dots$$

Jagrut Awaaz

$$\int \frac{1}{2x^2 + 3} dx = \dots + C$$



Jagrut Awaaz

Most Imp MCQs

62

$$\int \frac{x-1}{(x+1)^2} dx = \dots + C$$



Jagrut Awaaz

63

$$\int x^x (1 + \log x) dx = \dots + C$$

Jagrut Awaaz



Most Imp MCQs

64

If $|\bar{a} \times \bar{b}|^2 + |\bar{a} \cdot \bar{b}|^2 = 144$ and $|\bar{a}| = 4$ then $|\bar{b}| = \dots$

જો $|\bar{a} \times \bar{b}|^2 + |\bar{a} \cdot \bar{b}|^2 = 144$ અને $|\bar{a}| = 4$ તો $|\bar{b}| = \dots$

Jagrut Awaaz



Most Imp MCQs

65

If $\vec{a} = (1, 2, -3)$ and $\vec{b} = (-2, -4, -9)$ then unit vector of $\vec{a} - \vec{b}$ is:

જો $\vec{a} = (1, 2, -3)$ અને $\vec{b} = (-2, -4, -9)$ હોય તો $\vec{a} - \vec{b}$ નો એકમ સદિશ.....છે.

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$$\frac{\sin 50^\circ}{\sin 130^\circ} = \dots$$



Jagrut Awaaz

 $\cos 47^\circ \cos 13^\circ - \sin 47^\circ \sin 13^\circ = \dots$

Jagrut Awaaz

68

y intercept of line which passing through (2,2) and perpendicular to line $3x + y = 3$ is :

(2,2) માંથી પસાર થતી અને રેખા $3x + y = 3$ ને લંબ હોય તેવી રેખાનો y-અંતખંડ.....છે.

Jagrut Awaaz



Most Imp MCQs

69

The slope of line passing through point $(3,k)$ and $(2,7)$ is 2 then $k = \dots\dots\dots$

બિંદુ $(3,k)$ અને $(2,7)$ માંથી પસાર થતી રેખાનો ઢાળ 2 હોય તો $k = \dots\dots\dots$

Jagrut Awaaz

If the equation $\lambda x^2 + (2\lambda - 3)y^2 - 4x + 6y - 1 = 0$ is circle equation then centre of the circle is :

જો સમીકરણ $\lambda x^2 + (2\lambda - 3)y^2 - 4x + 6y - 1 = 0$ એ વર્તુળનું સમીકરણ હોય તો વર્તુળનું કેન્દ્ર.....છે.

Jagrut Awaaz



Most Imp MCQs

71

The median of observation 9, 15, 20, 22, 8, 17, 10, 18, 14, 25 is :

અવલોકનો 9, 15, 20, 22, 8, 17, 10, 18, 14, 25 નો મધ્યસ્થ.....છે.

Jagrut Awaaz

$$\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x} = \dots\dots\dots$$



Jagrut Awaaz

73

$$\lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)(2x - 3)}{2x^2 + x - 3} = \dots$$

Jagrut Awaaz



Most Imp MCQs

74

If $y = \frac{x}{a} + \frac{a}{x}$ then $\frac{dy}{dx} = \dots\dots\dots$

જલ $y = \frac{x}{a} + \frac{a}{x}$ ઢલ $\frac{dy}{dx} = \dots\dots\dots$

Jagrut Awaaz

Most Imp MCQs



75

$$\int_0^1 \frac{x}{1+x^2} dx = \dots$$

Jagrut Awaaz

If $A = \begin{bmatrix} 4 & 6 \\ -7 & 2 \end{bmatrix}$ then $\text{adj}(\text{adj} A) = \dots\dots\dots$

- (A) $-A$ (B) 1 (C) A (D) $.0$

Jagrut Awaaz

Main diagonal components (મુખ્ય વિકર્ણ ઘટકો) $\begin{bmatrix} 7 & 5 \\ 1 & 3 \end{bmatrix} = \dots\dots\dots$

- (A) (3,5) (B) (1,5) (C) (5,3) (D) (7,3)

If $\begin{bmatrix} x^2 - y^2 & 5 \\ 4 & x - y \end{bmatrix} = \begin{bmatrix} 16 & 5 \\ 4 & 2 \end{bmatrix}$ then find X and Y ?

(A) X=3, Y=5 (B) X=4, Y=2 (C) X=5, Y=3 (D) NOT POSSIBLE

$$A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix} \text{ find adj } A = \dots\dots\dots$$

- (A) $-A$ (B) I (C) A (D) A^T

Jagrut Awaaz

If $\tan\Theta = \frac{-4}{3}$, $\frac{\pi}{2} < \Theta < \pi$ then $\sec\Theta + \operatorname{cosec}\Theta = \dots\dots\dots$

(A) $3/5$ (B) $-3/5$ (C) $-5/12$ (D) $5/12$

Jagrut Awaaz

$$2\sin\left(A + \frac{\pi}{3}\right) = \dots\dots\dots$$

- (A) $\cos A + \sqrt{3}\sin A$ (B) 0 (C) $\sin A + 3\cos A$ (D) $\sin A + \sqrt{3}\cos A$

Jagrut Awaaz

$\sin A \cdot \sin (B-C) + \sin B \cdot \sin (C-A) + \sin C \cdot \sin (A-B) = \dots\dots\dots$

- (A) 0 (B) 1 (C) $\tan (A-B)$ (D) $\cot (A-B)$

Jagrut Awaaz

ચેક કરા, બળ $(1, 2, 3)$, $(-1, 2, 3)$ અને $(-1, 2, -3)$ ની અસર હેઠળ બિંદુ $(0, 1, -2)$ થી બિંદુ $(-1, 3, 2)$ સુધી સ્થાનાંતર કરે છે તો થયેલ કાર્ય શોધો.

Find the work done if a particle moves from point $(0, 1, -2)$ to point $(-1, 3, 2)$ under the action of forces $(1, 2, 3)$, $(-1, 2, 3)$ and $(-1, 2, -3)$.

(A) 21 (B) 12 (C) 25 (D) 52



If $\vec{x} \perp \vec{y}$ then $\vec{x} \cdot (\vec{x} + \vec{y}) = \dots\dots\dots$

(A) $|\vec{x}|$ (B) $|\vec{x}|^2$ (C) 0 (D) 1

Jagrut Awaaz

જો $A(5, 3)$, $B(p, -1)$, $C(4, -2)$ અને $D(-1, 1)$ તથા $AB \parallel CD$ હોય, તો p શોધો.

If $A(5, 3)$, $B(p, -1)$, $C(4, -2)$ and $D(-1, 1)$ and $AB \parallel CD$, then find p .

(A) $7/3$ (B) $13/3$ (C) $23/3$ (D) $35/3$

જો રેખાઓ $5x - my = 3$ અને $2x + 3y = 4$

(a) પરસ્પર સમાંતર હોય, તો m શોધો. (b) પરસ્પર લંબ હોય, તો m શોધો.

If the lines $5x - my = 3$ and $2x + 3y = 4$

(a) are parallel to each other, then find m . (b) are perpendicular to each other, then find m .

(A) $\frac{15}{2}, \frac{10}{3}$ (B) $\frac{15}{3}, \frac{10}{2}$ (C) $\frac{-15}{2}, \frac{-10}{3}$ (D) $\frac{-15}{2}, \frac{10}{3}$

$$\lim_{X \rightarrow 0} \frac{\tan X}{\sin X} = \dots\dots\dots$$

- (A) 0 (B) 1 (C) 2 (D) ∞



Jagrut Awaaz

13

$$\lim_{X \rightarrow 0} \frac{\sin(X-1)}{X^2-3X+2} = \dots\dots\dots$$

(A) -1 (B) 1 (C) 2 (D) -2

Jagrut Awaaz

If $X = a \sin \theta$ and $y = a (1 + \cos \theta)$ then $\frac{dy}{dx} = \dots\dots\dots$

- (A) $a \cos \theta$ (B) $-a \sin \theta$ (C) $-\tan \theta$ (D) $-\cot \theta$

If $Y = \log(e^{\sin x})$ then $\frac{dy}{dx} = \dots\dots\dots$

(A) $\sin x$ (B) $\cos x$ (C) $\tan x$ (D) $\cot x$

Jagrut Awaaz

If $Y = \log \sqrt{\frac{1+\sin x}{1-\sin x}}$ then $\frac{dy}{dx} = \dots\dots\dots$

(A) $\sin x$ (B) $\cos x$ (C) $\tan x$ (D) $\sec x$

Jagrut Awaaz

એક કણની ગતિનું સમીકરણ $s = t^3 - 6t^2 + 9t + 6$ છે, જ્યાં s મીટરમાં અને t સેકન્ડમાં છે. કણ જ્યારે પોતાની ગતિની દિશા બદલે ત્યારે s શોધો.

The equation of motion of a particle is $s = t^3 - 6t^2 + 9t + 6$, where s is in meters and t is in seconds. Find s when the particle changes its direction of motion.

(A) 10 & 6 m (B) 12 & 8 m (C) 10 & 8 m (D) 15 & 12m

$$\int_{-1}^1 \frac{x^3 - 1}{x - 1} \cdot dx = \dots\dots\dots$$

(A) $27/5$ (B) $27/6$ (C) $29/6$ (D) $35/8$

Jagrut Awaaz

$$\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx = \dots\dots\dots$$

- (A) $\sec x + \tan x$ (B) $-\cot x - \tan x$ (C) $\sin x - \cos x$ (D) $-\sin x + \cos x$

$$\int \log x \, dx = \dots\dots\dots$$

- (A) $1/x$ (B) $x \log x$ (C) $x \log x - x$ (D) x

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$$\int x \sin x \, dx = \dots\dots\dots$$

- (A) $\sec x + \tan x$ (B) $-x \cos x + \sin x$ (C) $x \sin x - \cos x$ (D) $-\sin x + \cos x$

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22

$$\log_{\sqrt{5}} \sqrt[3]{25} = \dots\dots\dots$$

- (A) 4/3 (B) 7/6 (C) 9/6 (D) 5/8

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$$4^{\log_2 5} = \dots\dots\dots$$

(A) 2 (B) 5 (C) 4 (D) 25 ✓

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$$\frac{\log x}{\log 8} = \frac{\log 256}{\log 64} \text{ then find } x = \dots\dots\dots$$

- (A) 8 (B) 16 (C) 32 (D) 64

આપેલી માહિતીનો મધ્યસ્થ શોધો : 2,-3,0,1,-5,-1,7,-8,-4.

Find the median of the given data: 2,-3,0,1,-5,-1,7,-8,-4

- (A) 0 (B) 1 (C) -1 (D) 2

$A = \begin{bmatrix} 5 & 3 \\ 4 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ then <u>AB</u> = ____.			
(a)	$\begin{bmatrix} 15 & 20 \\ 12 & 16 \end{bmatrix}$	(b)	$\begin{bmatrix} 13 & 14 \\ 9 & 7 \end{bmatrix}$
(c)	$\begin{bmatrix} 13 & 14 \\ 8 & 6 \end{bmatrix}$	(d)	$\begin{bmatrix} 15 & 20 \\ 9 & 7 \end{bmatrix}$

Jagrut Awaaz

For a Square matrix A, inverse of a matrix A does not exist if _____
ચોરસ શ્રેણિક માટે જો _____ હોય તો તેનો વ્યસ્ત ન મળે.

(a)	$ A = 0$	(b)	$ A \neq 0$
(c)	$A = A^T$	(d)	$A = -A^T$

Jagrut Awaaz

$$\begin{vmatrix} x & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 3 & 6 \end{vmatrix} = 0, \text{ હોય તો (then) } x = \underline{\hspace{1cm}}$$

(a)	1	(b)	2
(c)	0	(d)	3

Jagrut Awaaz

$A = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$ હોય તો (then) $A + \text{adj}(A) = \underline{\hspace{2cm}}$.			
(a)	$\begin{bmatrix} 5 & 3 \\ 0 & 1 \end{bmatrix}$	(b)	$\begin{bmatrix} 6 & -5 \\ 5 & 6 \end{bmatrix}$
(c)	$\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$	(d)	$\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

$$\sin 135^\circ + \cos 90^\circ = \underline{\hspace{2cm}}$$

(a)

1

(b)

 $\frac{1}{\sqrt{2}}$

(c)

 $\frac{1}{2}$

(d)

 $\sqrt{2}$

Jagrut Awaaz

$$\sin 3A = \underline{\hspace{2cm}}$$

(a) $3\sin A - 4\sin^3 A$

(b) $4\sin^3 A - 3\sin A$

(c) $4\sin A - 3\sin^3 A$

(d) $3\sin^4 A - 4\sin A$

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$\sin 40^\circ + \sin 20^\circ = \underline{\hspace{2cm}}$

(a) $\cos 10^\circ$

(b) $\sin 60^\circ$

(c) $\sin 20^\circ$

(d) $\cos 20^\circ$

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If $\vec{u} = 4i + 2j - k$ and $\vec{v} = 3j - i + mk$ are perpendicular than $m = \underline{\hspace{2cm}}$

જો $\vec{u} = 4i + 2j - k$ અને $\vec{v} = 3j - i + mk$ લંબ સદીશો હોય તો $m = \underline{\hspace{2cm}}$

A.	2	B.	-2
C.	12	D.	-10

$$(\bar{u} \times \bar{u}) + 3\bar{u} - \bar{u} = \underline{\hspace{1cm}}$$

A.

$4\bar{u}$

B.

0

C.

$3\bar{u}$

D.

$2\bar{u}$

રેખા $5x + y - 30 = 0$ નો ઢાળ ____ અને X -અંતઃખંડ ____ થશે.

For line $5x + y - 30 = 0$ slope of is ____ and X -intercept of the line is ____.

- | | | | | | | | |
|-----|------|-----|-------|-----|-------|-----|--------|
| (a) | 5, 6 | (b) | 5, -6 | (c) | -5, 6 | (d) | -5, -6 |
|-----|------|-----|-------|-----|-------|-----|--------|

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વર્તુળ $x^2 + y^2 = 144$ ની ત્રિજ્યા ____ થશે અને કેન્દ્ર ____ થશે .

Radius of the circle $x^2 + y^2 = 144$ is ____ and centre is ____.

(a)	144, (0,0)	(b)	12, (0,0)	(c)	12, (1,0)	(d)	12, (0,1)
-----	------------	-----	-----------	-----	-----------	-----	-----------

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$g(x) = 6^x + \log(\cos x)$ હોય તો (then) $g(0) = \underline{\hspace{1cm}}$.

(a)	6	(b)	2
(c)	1	(d)	7

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$$\lim_{x \rightarrow 0} \left(\frac{3x + 14}{7} + \frac{\tan x}{x} \right) = \underline{\hspace{2cm}}$$

(a)	0	(b)	7
(c)	2	(d)	3

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$$\lim_{n \rightarrow \infty} \frac{2n^2 - 4n + 3}{3n^2 + 3n + 2}$$

(a)	0	(b)	3/2
(c)	2/3	(d)	-4/3

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$$\frac{d}{dx} [x \sin x + 11] = \underline{\hspace{2cm}}$$

- | | | | | | | | |
|-----|-------------------|-----|----|-----|-----------------|-----|-----------|
| (a) | <u>cos</u> x + 11 | (b) | 11 | (c) | x cos x + sin x | (d) | x + sin x |
|-----|-------------------|-----|----|-----|-----------------|-----|-----------|

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$$\frac{d}{dx} \left(\frac{2x^3 - 3x^2 - 3}{x} \right)$$

(a)

$$4x - 3 + 3x^{-2}$$

(b)

$$4x + 6x^{-2}$$

(c)

$$4x - 6x^{-2}$$

(d)

$$4x - 3 - 3x^{-2}$$

$y = \log(5x)$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

(a) $\frac{1}{5}$

(b) $\frac{5}{x}$

(c) $\frac{1}{5x}$

(d) $\frac{1}{x}$

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If $x = \sec \theta + \tan \theta$ and $y = \sec \theta - \tan \theta$, then $\frac{dy}{dx} =$ _____.

A. $\frac{-y}{x}$

B. $\frac{-1}{x^2}$

C. $\frac{x}{y}$

D. 0

$$\int f(x)dx = \log|\sin x| + c \text{ તો } f(x) = \underline{\hspace{2cm}}.$$

A. $\frac{1}{\sin x}$

B. $\frac{\cos x}{\sin x}$

C. $\log(\cos x)$

D. એક પણ નહીં

$$\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx = \text{_____} + c$$

a

$$\frac{x^2}{2} + 2x + \log |x|$$

b

$$\frac{x^2}{2} + \log |x|$$

c

$$\frac{1}{2\sqrt{x}} + x$$

d

$$\frac{1}{2\sqrt{x}} + \log |x|$$

A solid yellow square is located in the top left corner of the slide.
$$\int \log x \, dx$$

A. $x \log x + c$

B. $x \log x - x + c$

C. $\log|\cos x| + c$

D. *none of these*

A solid yellow square is located in the top-left corner of the page.
$$\int_{-1}^1 x^3 dx = \underline{\hspace{2cm}}$$

a	1	b	0
c	2	d	3

$$\log_{23} 1 + \log_a 2 + \log_a 3 = \underline{\hspace{2cm}}$$

(a)

a

(b) $\log_a 5$ **(c)** $\log_a 7$ **(d)** $\log_a 6$

$$\log (4) + \log \left(\frac{1}{4} \right) + \log_2 \sqrt{2} = \underline{\hspace{1cm}}$$

(a)	$\frac{1}{\sqrt{2}}$	(b)	$\frac{1}{2}$
(c)	0	(d)	1

Mean of first six odd numbers is ____.

પ્રથમ છ અચુગ્મ સંખ્યાઓનો મધ્યક __ થાય.

(a)	5	(b)	7
(c)	5.5	(d)	6

76

$$\int \frac{x^2 + 5x + 6}{x^2 + 3x} dx = \dots + C$$

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77

$$\int e^x x^2 (x + 3) dx = \dots + C$$

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Most Imp MCQs

If $\log_x y = 100$ and $\log_2 x = 10$ then $y = \dots\dots\dots$
જો $\log_x y = 100$ અને $\log_2 x = 10$ તો $y = \dots\dots\dots$

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79

$$\frac{d}{dx} \ln(\cos x) = \dots$$

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Most Imp MCQs

80

$$\int_{-4}^4 \sin^5 x \, dx = \dots$$



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Most Imp MCQs

81

Equation of line parallel to $y = \frac{-3x}{5} + 6$ is

રેખા $y = \frac{-3x}{5} + 6$ ને સમાંતર રેખાનું સમીકરણ..... છે.

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Most Imp MCQs

82

The mode of observation 3, 5, 7, 7, 9, 3, 3 is

અવલોકનો 3, 5, 7, 7, 9, 3, 3 નો બહુલક..... છે.

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83

$$\frac{d}{dx}(\sin x \cdot \tan x) = \dots\dots\dots$$

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Most Imp MCQs

For the matrix M and N if $M + N = \begin{bmatrix} 5 & 6 \\ 4 & 2 \end{bmatrix}$ and

$M - N = \begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix}$ then

શ્રેણિક M અને N માટે જો $M + N = \begin{bmatrix} 5 & 6 \\ 4 & 2 \end{bmatrix}$ અને

$M - N = \begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix}$ તો

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Most Imp MCQs

85

The angle between line $x - 3y + 1 = 0$ and the x -axis is

.....

રેખા $x - 3y + 1 = 0$ અને x -અક્ષ વચ્ચેનો ખૂણો..... છે.

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Most Imp MCQs

If $\log_2 a = 3$ and $\log_2 b = 2$ then $\log_2 (ab) = \dots\dots\dots$
જો $\log_2 a = 3$ અને $\log_2 b = 2$ તો $\log_2 (ab) = \dots\dots\dots$

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Most Imp MCQs

87

The equation for the circle with centre (1,2) and radius of $\sqrt{5}$ is....

કેન્દ્ર (1,2) અને ત્રિજ્યા $\sqrt{5}$ હોય તેવા વર્તુળનું સમીકરણ..... છે.

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$$\lim_{x \rightarrow 5} \frac{32x - 160}{x^2 - 5x} = \dots$$



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Most Imp MCQs

89

Period of $\sin \frac{x}{4} + \cos \frac{x}{10}$ is

$\sin \frac{x}{4} + \cos \frac{x}{10}$ નું આવર્તમાન છે.

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Most Imp MCQs

90

If $\begin{bmatrix} a+b & c \\ a-2b & c+2d \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix}$ then

$$a + b + c + d = \dots\dots$$

જો $\begin{bmatrix} a+b & c \\ a-2b & c+2d \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix}$ તો

$$a + b + c + d = \dots\dots$$

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Most Imp MCQs

91

If $\sin x = \frac{1}{2}$ then $\sin 2x = \dots\dots$

જો $\sin x = \frac{1}{2}$ તો $\sin 2x = \dots\dots$

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$$\lim_{x \rightarrow \infty} \frac{x^2 + \sqrt{2}x + 5}{x^2 - 4x} = \dots\dots$$



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93

$$2\sin 7\theta \cos 3\theta = \dots\dots$$

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Most Imp MCQs

94

If A $(-1, -2, 3)$ and B $(1, 2, -1)$ then direction cosine of $\overrightarrow{AB}$ is.....

જો A $(-1, -2, 3)$ અને B $(1, 2, -1)$ તો $\overrightarrow{AB}$ ની દિક્કોસાઈનછે.

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Most Imp MCQs

95

If $3\hat{i} + 4\hat{j} + 9\hat{k}$ and $a\hat{i} - 3\hat{j} + \hat{k}$ are perpendicular then a
=

જો $3\hat{i} + 4\hat{j} + 9\hat{k}$ અને $a\hat{i} - 3\hat{j} + \hat{k}$ પરસ્પર લંબ હોય તો $a =$
....

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96

$$\lim_{x \rightarrow 0} \frac{e^{3+x} - \sin x - e^3}{x} = \dots\dots$$

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Most Imp MCQs

97

If $\begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$ then sum of all possible value of a is :

જો $\begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$ તો a ની શક્ય તમામ કિંમતોનો
સરવાળો.....છે.

Jagrut Awaaz



Most Imp MCQs

98

If $f(x) = x^3 + 4^x + 2^3$ then $f'(0) = \dots\dots\dots$
જલ $f(x) = x^3 + 4^x + 2^3$ ઢલ $f'(0) = \dots\dots\dots$

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Most Imp MCQs

99

System of equation $4x + 2y = 7$, $6x + 3y = 9$ has
સમીકરણ સંહતી $4x + 2y = 7$, $6x + 3y = 9$ ને.....

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100

If $f(x) = 8 - 2x$ then $f^{-1}(1) = \dots\dots\dots$
જલ $f(x) = 8 - 2x$ ંલ $f^{-1}(1) = \dots\dots\dots$

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Most Imp MCQs

101

If $\log_4(\log_5(\log_6 x)) = 2$ then $x = \dots\dots\dots$

જો $\log_4(\log_5(\log_6 x)) = 2$ તો $x = \dots\dots\dots$

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Most Imp MCQs

102

$$\frac{d}{dx} e^{x^2} = \dots\dots$$



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103

$$\cos\left(\frac{\pi}{2} + \sin^{-1}\frac{3}{8}\right) = \dots\dots\dots$$

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104

$$\int x e^{4x} dx = \dots\dots\dots + C$$

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105

$$\frac{d}{dx}(\sin x^3 \cdot \cos x^2) = \dots$$

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Most Imp MCQs

106

Equation $x = 3 \cos \theta$ and $y = 3 \sin \theta$ represents the equations is:

સમીકરણો $x = 3 \cos \theta$ અને $y = 3 \sin \theta$ દ્વારા દર્શાવાતું સમીકરણ છે.

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107

$$\int_{-2}^2 \mathbf{x^4 dx} = \dots\dots$$

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$$\int \frac{4}{x\sqrt{16x^2 - 144}} dx = \dots + C$$



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Most Imp MCQs

109

Period of $\cos\left(\frac{2\pi x}{3}\right)$ is

$\cos\left(\frac{2\pi x}{3}\right)$ નું આવર્તમાન છે.

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Most Imp MCQs

110

Centre of the circle $3x^2 + 3y^2 - 6x + y - 9 = 0$ is :
ગાંઠી $3x^2 + 3y^2 - 6x + y - 9 = 0$ નું કેન્દ્ર..... છે.

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Most Imp MCQs

111

If $\frac{dy}{dx} = 4x^3 - 3x^2 + 7$ and $y = 30$ when $x = 2$ then

$y = \dots\dots\dots$

જો $\frac{dy}{dx} = 4x^3 - 3x^2 + 7$ અને $y = 30$ જ્યારે $x = 2$ તો

$y = \dots\dots\dots$

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Most Imp MCQs

112

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ then $AB = \dots$

જો $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ અને $B = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ તો $AB = \dots$

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Most Imp MCQs

113

If $\log_{10} 5 + \log_{10} (5x+1) = \log_{10} (x+5) + 1$ then $x = \dots$
જલ $\log_{10} 5 + \log_{10} (5x+1) = \log_{10} (x+5) + 1$ ઢલ $x = \dots$

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Most Imp MCQs

114

The equation of line passing through (2,8) and (4,15) is :
બિંદુ (2,8) અને (4,15) માંથી પસાર થતી રેખાનું
સમીકરણ..... છે.

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Most Imp MCQs

115

If $y = x^{e^x}$ then $\frac{dy}{dx} = \dots\dots\dots$

જલ $y = x^{e^x}$ ંલ $\frac{dy}{dx} = \dots\dots\dots$

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Most Imp MCQs

116

If $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$ then

જલ $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$ નલ

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Most Imp MCQs

If $\begin{vmatrix} x & 4 & 10 \\ 5 & 2 & 5 \\ 7 & 3 & x \end{vmatrix} = 0$ then $x = \dots\dots$

જલ $\begin{vmatrix} x & 4 & 10 \\ 5 & 2 & 5 \\ 7 & 3 & x \end{vmatrix} = 0$ તલ $x = \dots\dots$

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118

$$\frac{\sin 5x + \sin 3x}{\cos 5x + \cos 3x} = \dots$$

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Most Imp MCQs

119

If $|\bar{a}| = 5$, $|\bar{b}| = 3$ and $\bar{a} \cdot \bar{b} = 4$ then $|\bar{a} - \bar{b}| = \dots\dots\dots$

જો $|\bar{a}| = 5$, $|\bar{b}| = 3$ અને $\bar{a} \cdot \bar{b} = 4$ તો $|\bar{a} - \bar{b}| = \dots\dots\dots$

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Most Imp MCQs

If $(\hat{i} + x\hat{j} + y\hat{k}) \times (2\hat{i} + 6\hat{j} + 27\hat{k}) = \vec{0}$ then $x + y = \dots$

જલ $(\hat{i} + x\hat{j} + y\hat{k}) \times (2\hat{i} + 6\hat{j} + 27\hat{k}) = \vec{0}$ તલ $x + y = \dots$

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Most Imp MCQs

121

If mean of 3, 4, 5, 6, x, 7, 10 is 7 then $x = \dots\dots$

જો 3, 4, 5, 6, x, 7, 10 નો મધ્યક 7 હોય તો $x = \dots\dots$

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$$\lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x + x^2 \cos^2 x} = \dots$$



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$$\lim_{x \rightarrow 0} \frac{15^x - 5^x - 3^x + 1}{x^2} = \dots$$



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$$\lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3} = \dots$$



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Most Imp MCQs

125

If $f(x) = \log\left(\frac{x+5}{x^2-1}\right) + \sin(15x)^\circ$ then $f(3) = \dots\dots\dots$

જો $f(x) = \log\left(\frac{x+5}{x^2-1}\right) + \sin(15x)^\circ$ તો $f(3) = \dots\dots\dots$

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